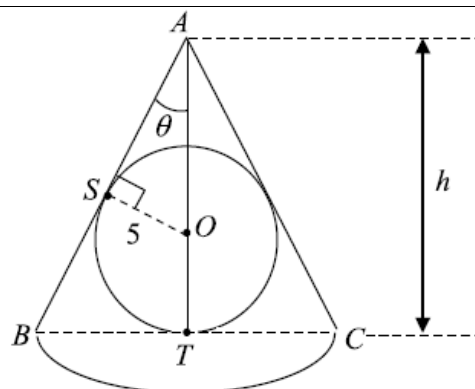


1	ACSB/2025/I/06
	The function f is defined by $f(x) = \frac{x^2 + 4}{x^2 - 1}, x \neq \pm 1$. (a) Explain, with working, whether f is an increasing or a decreasing function for $x > 1$. [4] (b) A point Q moves along the curve $y = f(x)$ in such a way that the y-coordinate of Q is increasing at a rate of 0.1 units per second. Find the rate of change of the x-coordinate of Q when $x = 2$. [2]
2	ACSB/2025/II/06
	A closed can, in the shape of a cylinder, contains 600 cm^3 of liquid when full. The cylinder, of radius $r \text{ cm}$ and height $h \text{ cm}$, is made from a thin sheet of metal. The total external surface area of cylinder is $A \text{ cm}^2$. (a) Show that $A = 2\pi r^2 + \frac{1200}{r}$. [3] (b) Given that r can vary, find the stationary value of A and determine its nature. [5]
3	BASS/2025/I/08
	A function $g(x)$ is given by $g(x) = -2x^3 + mx^2 + nx + 2$ where m and n are constants. (a) Show that if $g(x)$ is always decreasing, then $m^2 < -6n$. [4] (b) Explain why $y = \frac{10}{5-x} - 2$ does not have a stationary point. [2]
4	BASS/2025/II/05
	(a) It is given that $x = e^{-2t} + 3e^{-t}$. Show that $\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 2x = 0$. [3] (b) Solve $2^{2x-1} - 11(2^{x-2}) + 3 = 0$. [4]



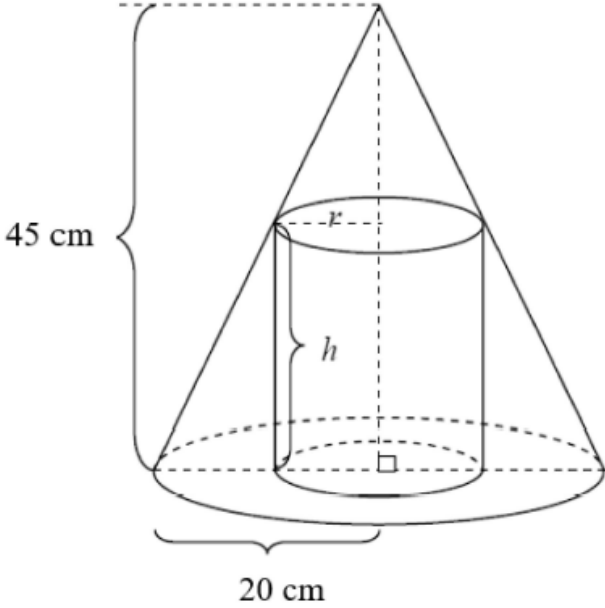
A sphere, with centre O and radius 5 cm, is inscribed in a cone with vertical height h cm such that the sphere is in contact with the curve surface and the base of the cone. S and T are points of contact between the sphere and cone. θ is the angle between the slant height AB and the vertical height AT of the cone.

(a) Show that $\tan \theta = \frac{5}{\sqrt{h^2 - 10h}}$. [2]

(b) Show the volume of the cone, $V \text{ cm}^3$, is given by $V = \frac{25\pi h^2}{3(h-10)}$. [3]

(c) Given that h can vary, find the value of h for which the volume is stationary. [4]

(d) Determine whether this value of h gives a maximum or minimum value for the volume of the cone. [2]

6	CCHY/2025/I/7
	The diagram shows a solid cylinder of radius r cm and height h cm inscribed in a hollow cone of height 45 cm and base radius 20 cm. The cylinder rests on the base of the cone and the circumference of the top surface of the cylinder touches the curved surface of the cone.  (i) Show that the volume, $V \text{ cm}^3$, of the cylinder is given by $V = 45\pi r^2 - \frac{9}{4}\pi r^3$. [3] (ii) Given that r can vary, find the maximum volume of the cylinder, leaving your answer in terms of π. [5] (iii) Hence show that the cylinder occupies at most $\frac{4}{9}$ of the volume of the cone. [2]
7	CCHY/2025/II/3
	(a) Show that $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$. [2] (b) A curve has equation $y = e^{-ax} \cot x$, where a is a positive constant. There is exactly one point in the interval $-\frac{1}{2}\pi < x < 0$ at which the tangent is parallel to the x-axis. Find the value of a and state the exact x-coordinate of this point. [5]

8	CCHY/2025/II/11
	(a) A curve $y = f(x)$ is such that $f'(x) = ax^2 - 6x + b$, where a and b are constants. (i) Given that the curve is always increasing, what conditions must apply to a and b? [3] It is now given that $a = 5$. (ii) The curve intersects the y-axis at $(0, 4)$ and passes through the points $(-3, -80)$ and $(3, k)$. Find the value of k. [5] This question is not related to part (a). (b) It is given that $y = g(x)$ is a solution of the equation $e^{-2x} \left(\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + y \right) = -e^k$, where k is a constant. The point $(1, 0)$ is a stationary point on the curve $y = g(x)$. Find the nature of this stationary point. [3]
9	GMS/2025/I/1
	The equation of a curve is $y = \frac{3}{x^2} + \frac{4}{3x-5}$. Given that x is decreasing at a constant rate of 0.4 units per second, find the rate of change of y when $x = 2$. [4]
10	GMS/2025/I/8
	The point A lies on the curve $y = x \ln x$. The tangent to the curve at A is parallel to the line $y + 3 = 2x$. (a) Find the coordinates of A. [4] The normal to the curve at A meets the line $y + 3 = 2x$ at the point B. (b) Show that the x-coordinate of B is $k(e+2)$, where k is a constant to be found. [3]
11	GMS/2025/I/13
	A curve has the equation $y = \frac{e^{2x}}{x+2}$, where $x \neq -2$. (a) Find $\frac{dy}{dx}$. [3] (b) Show that the coordinates of the turning point are $\left(-\frac{3}{2}, \frac{2}{e^3}\right)$. [4] (c) By considering the sign of $\frac{dy}{dx}$, determine whether the turning point is a maximum or a minimum. [3]

12	GMS/2025/II/3
	Two flower beds are planned for a large garden. One is in the shape of a circle of radius r m and the other is a square of side x m. The total perimeter of the circle and the square is 40 m. (a) Express x in terms of r and π. [2] (b) The combined area of the two flower beds is A m². Show that $A = \frac{\pi}{4}(4 + \pi)r^2 - 10\pi r + 100.$ [3] Given that r may vary, (c) find the value of r for which A is stationary, correct to 3 significant figures, [3] (d) find the stationary value of A, (e) determine whether this stationary value of A is a maximum or a minimum. [2]
13	GMS/2025/II/2
	(a) Prove that $\operatorname{cosec} 2\theta + \cot 2\theta = \cot \theta$. [3] (b) Hence find all the angles between $-\pi$ and π which satisfy the equation $\operatorname{cosec} 2\theta + \cot 2\theta + \frac{1}{\sqrt{3}} = 0.$ [3] (c) It is given that $f'(x) = \sin 2x + \cos 3x$ and $f\left(\frac{\pi}{6}\right) = 0$. Show that $f''(x) + 9f(x) = -\frac{3}{4} - \frac{5}{2}\cos 2x$. [5]

1	(a)	$f'(x) = \frac{(x^2-1)(2x) - (x^2+4)(2x)}{(x^2-1)^2}$ $= \frac{2x^3 - 2x - 2x^3 - 8x}{(x^2-1)^2}$ $= \frac{-10x}{(x^2-1)^2}$ for $x > 1$, $-10x < 0$ and $(x^2-1)^2 > 0$ and hence $f'(x) < 0$. <u>This implies that f is an decreasing function.</u>
	(b)	$\frac{dx}{dt} = \frac{dx}{dy} \times \frac{dy}{dt}$ $\frac{dx}{dt} = \frac{(4-1)^2}{-10(2)} \times 0.1$ $= -0.045 \text{ units/sec or } \frac{-9}{200}$
2	(a)	$h = \frac{600}{\pi r^2}$ $A = 2\pi r^2 + 2\pi r \times \frac{600}{\pi r^2}$ $= 2\pi r^2 + \frac{1200}{r}$ $\frac{dA}{dr} = 4\pi r - \frac{1200}{r^2}$ for stationary value, $\frac{dA}{dr} = 4\pi r - \frac{1200}{r^2} = 0$ $r^3 = \frac{300}{\pi}$ $r = 4.57078$ $A = 2\pi (4.57078)^2 + \frac{1200}{4.57078}$
	(b)	$= 394 \text{ cm}^2$ $\frac{d^2A}{dr^2} = 4\pi + \frac{2400}{r^3}$ when $r = 4.57078$, $\frac{d^2A}{dr^2} = 4\pi + \frac{2400}{4.57078^3} = 38.77 > 0$ $\Rightarrow$ the stationary value is minimum

3	a $g(x) = -2x^3 + mx^2 + nx + 2$ $g'(x) = -6x^2 + 2mx + n$ For $g(x)$ to be always decreasing $g'(x) < 0$ for all x $-6x^2 + 2mx + n < 0$ For $-6x^2 + 2mx + n < 0$, discriminant < 0 $-6x^2 + 2mx + n < 0$ $(2m)^2 - 4(-6)(n) < 0$ $4m^2 + 24n < 0$ $4m^2 < -24n$ $m^2 < -6n \text{ (shown)}$ b $y = \frac{10}{5-x} - 2$ $\frac{dy}{dx} = \frac{10}{(5-x)^2}$ Since $(5-x)^2 > 0$ for all real values of x where $x \neq 5$ $\frac{dy}{dx} = \frac{10}{(5-x)^2} > 0 \text{ and } \frac{dy}{dx} \neq 0,$ Therefore the curve has no turning points
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4**a**

$$x = e^{-2t} + 3e^{-t}$$

$$\begin{aligned}\frac{dx}{dt} &= -2e^{-2t} + (-3e^{-t}) \\ &= -2e^{-2t} - 3e^{-t}\end{aligned}$$

$$\begin{aligned}\frac{d^2x}{dt^2} &= \frac{d}{dt}(-2e^{-2t} - 3e^{-t}) \\ &= (-2)(-2e^{-2t}) - (-3e^{-t}) \\ &= 4e^{-2t} + 3e^{-t}\end{aligned}$$

$$\begin{aligned}\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 2x &= 4e^{-2t} + 3e^{-t} + 3(-2e^{-2t} - 3e^{-t}) + 2(e^{-2t} + 3e^{-t}) \\ &= 4e^{-2t} + 3e^{-t} - 6e^{-2t} - 9e^{-t} + 2e^{-2t} + 6e^{-t} \\ &= (4e^{-2t} - 6e^{-2t} + 2e^{-2t}) + (3e^{-t} - 9e^{-t} + 6e^{-t}) \\ &= 0 \quad (\text{proven})\end{aligned}$$

b

$$2^{2x} \times \frac{1}{2} - 11(2^x \times 2^{-2}) + 3 = 0$$

$$\text{Let } y = 2^x$$

$$\frac{1}{2}y^2 - \frac{11}{4}y + 3 = 0$$

$$2y^2 - 11y + 12 = 0$$

$$(2y - 3)(y - 4) = 0$$

$$y = \frac{3}{2} \quad \text{or} \quad 4$$

$$2^x = \frac{3}{2} \quad \text{or} \quad 4$$

$$x = \frac{\lg \frac{3}{2}}{\lg 2} \quad \text{or} \quad 2$$

$$\therefore x = 0.585, 2$$

5

a Since $AT = h$ and $OT = 5$, therefore, $AO = h - 5$

$$AS^2 + 5^2 = (h - 5)^2$$

$$AS^2 + 25 = h^2 - 10h + 25$$

$$AS = \sqrt{h^2 - 10h}$$

In $\triangle ASO$,

$$\tan \theta = \frac{5}{AS}$$

$$= \frac{5}{\sqrt{h^2 - 10h}} \quad (\text{Shown})$$

b In $\triangle ABT$,

$$\tan \theta = \frac{BT}{h}$$

$$BT = h \tan \theta$$

$$= \frac{5h}{\sqrt{h^2 - 10h}} \quad [\text{from (a)}]$$

$$V = \frac{1}{3} \times \pi \times BT^2 \times h$$

$$= \frac{1}{3} \times \pi \times \left(\frac{5h}{\sqrt{h^2 - 10h}} \right)^2 \times h$$

$$= \frac{1}{3} \times \pi \times \frac{25h^2}{h(h - 10)} \times h$$

$$= \frac{25\pi h^2}{3(h - 10)} \quad (\text{Shown})$$

c $\frac{dV}{dh} = \frac{(3h - 30)(50\pi h) - 25\pi h^2(3)}{(3h - 30)^2}$

$$= \frac{150\pi h^2 - 1500\pi h - 75\pi h^2}{(3h - 30)^2}$$

$$= \frac{75\pi h^2 - 1500\pi h}{(3h - 30)^2}$$

$$\frac{75\pi h^2 - 1500\pi h}{(3h - 30)^2} = 0$$

$$75\pi h(h - 20) = 0$$

$$h = 0 \quad (\text{reject}) \quad \text{or} \quad 20$$

d **Method 1:**

h	19.9	20	20.1
$\frac{dV}{dh}$	< 0	0	> 0
Slope	\	—	/

By 1st derivative test, V is minimum

Method 2:

$$\frac{dV}{dh} = \frac{75\pi h^2 - 1500\pi h}{(3h - 30)^2}$$

$$\frac{d^2V}{dh^2} = \frac{(3h - 30)^2(150\pi h - 1500\pi) - (75\pi h^2 - 1500\pi h)(6)(3h - 30)}{(3h - 30)^4}$$

$$\frac{d^2V}{dh^2} \Big|_{h=20} = \frac{5}{3}\pi > 0$$

By 2nd derivative test, V is minimum

6

(i)

$$\frac{45-h}{45} = \frac{r}{20}$$

$$20(45-h) = 45r$$

$$900 - 20h = 45r$$

$$h = \frac{900 - 45r}{20}$$

$$= 45 - \frac{9}{4}r$$

$$V = \pi r^2 h$$

$$= \pi r^2 \left(45 - \frac{9}{4}r\right)$$

$$= 45\pi r^2 - \frac{9}{4}\pi r^3 \text{ (shown)}$$

(ii)

$$\frac{dV}{dr} = 90\pi r - \frac{27}{4}\pi r^2$$

when V is max,

$$90\pi r - \frac{27}{4}\pi r^2 = 0$$

$$\pi r \left(90 - \frac{27}{4}r\right) = 0$$

$$r = 0 \quad \text{or} \quad r = 13\frac{1}{3}$$

(rejected)

$$\frac{d^2V}{dr^2} = 90\pi - \frac{27}{2}\pi r$$

$$\text{when } r = \frac{40}{3},$$

$$\frac{d^2V}{dr^2} = 90\pi - \frac{27}{2}\pi \left(\frac{40}{3}\right)$$

$$= -282.74$$

$$< 0$$

Since $\frac{d^2V}{dr^2} < 0$, $V = \frac{8000\pi}{3} \text{ cm}^3$ is a maximum volume.

(iii)

$$\text{Volume of cone} = \frac{1}{3}\pi(20)^2(45)$$

$$= 6000\pi$$

$$\therefore \text{ratio} = \frac{\frac{8000\pi}{3}}{6000\pi}$$

$$= \frac{4}{9} \text{ (shown)}$$

(a)

$$\begin{aligned}
 & \frac{d}{dx}(\cot x) \\
 &= \frac{d}{dx}\left(\frac{\cos x}{\sin x}\right) \\
 &= \frac{\sin x(-\sin x) - \cos x(\cos x)}{\sin^2 x} \\
 &= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} \\
 &= \frac{-1}{\sin^2 x} \quad \text{or} \quad -1 - \cot^2 x \\
 &= -\operatorname{cosec}^2 x
 \end{aligned}$$

(b)

$$\begin{aligned}
 \frac{dy}{dx} &= e^{-ax}(-\operatorname{cosec}^2 x) - ae^{-ax} \cot x \\
 &= -e^{-ax}(\operatorname{cosec}^2 x + a \cot x)
 \end{aligned}$$

The tangent is parallel to x -axis means $\frac{dy}{dx} = 0$.

$$-e^{-ax}(\operatorname{cosec}^2 x + a \cot x) = 0$$

Since $e^{-ax} > 0$ for all values of x ,

$$-(\operatorname{cosec}^2 x + a \cot x) = 0$$

$$\operatorname{cosec}^2 x + a \cot x = 0$$

$$\cot^2 x + 1 + a \cot x = 0$$

Since there is only 1 point, the above equation only has 1 solution, i.e. discriminant = 0.

$$a^2 - 4(1)(1) = 0$$

$$a = \pm 2$$

$$a = 2 \quad (a > 0)$$

$$\cot^2 x + 2 \cot x + 1 = 0$$

$$(\cot x + 1)^2 = 0$$

$$\cot x = -1$$

$$\tan x = -1$$

$$x = -\frac{\pi}{4}$$

$$\begin{aligned}
 & \frac{d}{dx}(\cot x) \\
 &= \frac{d}{dx}\left(\frac{1}{\tan x}\right) = \frac{d}{dx}(\tan x)^{-1} \\
 &= -1(\tan x)^{-2}(\sec^2 x) \\
 &= -\frac{1}{\tan^2 x \cos^2 x} \\
 &= \frac{-1}{\sin^2 x} \\
 &= -\operatorname{cosec}^2 x
 \end{aligned}$$

(a)

For the curve to be always increasing, i.e. $f'(x) > 0$, that is

$$ax^2 - 6x + b > 0$$

This means that there are no roots, i.e. discriminant < 0 .

$$(-6)^2 - 4(a)(b) < 0$$

$$36 - 4ab < 0$$

$$36 < 4ab$$

$$ab > 9$$

$$\text{and } a > 0$$

(b)

$$\frac{dy}{dx} = 5x^2 - 6x + b$$

$$y = \frac{5}{3}x^3 - 3x^2 + bx + 4$$

When $x = -3$, $y = 80$.

$$80 = \frac{5}{3}(-3)^3 - 3(-3)^2 + b(-3) + 4$$

$$-80 = -45 - 27 - 3b + 4$$

$$3b = 12$$

$$b = 4$$

When $x = 3$, $y = k$.

$$k = \frac{5}{3}(3)^3 - 3(3)^2 + 4(3) + 4$$

$$= 45 - 27 + 12 + 4$$

$$= 34$$

(c)

Since it is a stationary point, $\frac{dy}{dx} = 0$. Together with $(1, 0)$,

we can substitute $x = 1$, $y = 0$, $\frac{dy}{dx} = 0$. The equation is

$$\text{now reduced to } e^{-2(1)} \left(\frac{d^2y}{dx^2} - 2(0) + 0 \right) = -e^k$$

$$e^{-2} \left(\frac{d^2y}{dx^2} \right) = -e^k$$

$$\frac{d^2y}{dx^2} = -e^{2+k}$$

Since exponential functions are always positive, we have $-e^{2+k}$ to be negative.

Since $\frac{d^2y}{dx^2} < 0$, by 2nd derivative test, $(1, 0)$ is a maximum point.

9

Given: $y = \frac{3}{x^2} + \frac{4}{3x-5}$

$$= 3x^{-2} + 4(3x-5)^{-1}$$

$$\frac{dy}{dx} = -\frac{6}{x^3} + 4(-1)(3x-5)^{-2}(3)$$

$$= -\frac{6}{x^3} - \frac{12}{(3x-5)^2}$$

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

When $x = 2$ $\frac{dy}{dt} = \left[-\frac{6}{8} - \frac{12}{1} \right] (-0.4)$
 $= 5.1$

(a)

$$y = x \ln x$$

$$\frac{dy}{dx} = x \left(\frac{1}{x} \right) + \ln x$$

$$= 1 + \ln x$$

Tangent is parallel to $y + 3 = 2x$

$$1 + \ln x = 2$$

$$\ln x = 1$$

$$x = e,$$

$$y = e \ln e$$

$$= e$$

A is (e, e)

(b)

$$\text{Gradient of normal} = -\frac{1}{2}$$

$$\text{Equation of normal: } y - e = -\frac{1}{2}(x - e)$$

$$y = -\frac{1}{2}x + \frac{3}{2}e$$

$$\text{At } B, \quad -\frac{1}{2}x + \frac{3}{2}e = 2x - 3$$

$$\frac{5}{2}x = \frac{3}{2}e + 3$$

$$x = \frac{2}{5} \left(\frac{3}{2}e + 3 \right)$$

$$= \frac{3}{5}e + \frac{6}{5}$$

$$= \frac{3}{5}(e + 2)$$

(a)

$$\begin{aligned}
 y &= \frac{e^{2x}}{x+2} \\
 \frac{dy}{dx} &= \frac{(x+2)(2e^{2x}) - (e^{2x})}{(x+2)^2} \\
 &= \frac{(e^{2x})[2x+4-1]}{(x+2)^2} \\
 &= \frac{e^{2x}(2x+3)}{(x+2)^2}
 \end{aligned}$$

(b)

At turning point, $\frac{dy}{dx} = 0$

$$\begin{aligned}
 \frac{e^{2x}(2x+3)}{(x+2)^2} &= 0 \\
 e^{2x}(2x+3) &= 0
 \end{aligned}$$

Since $e^{2x} \neq 0$, $2x+3 = 0$

$$\begin{aligned}
 x &= -\frac{3}{2} \\
 y &= \frac{e^{2\left(-\frac{3}{2}\right)}}{\left(-\frac{3}{2}+2\right)} \\
 &= \frac{2}{e^3}
 \end{aligned}$$

The turning point is $\left(-\frac{3}{2}, \frac{2}{e^3}\right)$

(c)

$$\frac{dy}{dx} = \frac{e^{2x}(2x+3)}{(x+2)^2}$$

At turning point, $x = -\frac{3}{2}$.

When $x < -\frac{3}{2}$, $\frac{dy}{dx} = \frac{e^{2x}(2x+3)}{(x+2)^2} < 0$

When $x = -\frac{3}{2}$, $\frac{dy}{dx} = \frac{e^{2x}(2x+3)}{(x+2)^2} = 0$

When $x > -\frac{3}{2}$, $\frac{dy}{dx} = \frac{e^{2x}(2x+3)}{(x+2)^2} > 0$

$\left(-\frac{3}{2}, \frac{2}{e^3}\right)$ is a minimum point.

(a)

$$4x + 2\pi r = 40$$

$$x = \frac{40 - 2\pi r}{4}$$

$$= \frac{20 - \pi r}{2}$$

(b)

$$\text{Area} = x^2 + \pi r^2$$

$$= \left(\frac{20 - \pi r}{2} \right)^2 + \pi r^2$$

$$= \frac{400 - 40\pi r + \pi^2 r^2}{4} + \pi r^2$$

$$= 100 - 10\pi r + \frac{1}{4}\pi^2 r^2 + \pi r^2$$

$$= 100 - 10\pi r + \frac{1}{4}\pi r^2 (\pi + 4)$$

$$= \frac{\pi}{4}(4 + \pi)r^2 - 10\pi r + 100 \quad (\text{Shown})$$

(c)

$$A = \frac{\pi}{4}(4 + \pi)r^2 - 10\pi r + 100$$

$$\frac{dA}{dr} = \frac{\pi}{4}(4 + \pi)(2r) - 10\pi$$

When A is stationary, $\frac{\pi}{4}(4 + \pi)(2r) - 10\pi = 0$

$$2r = \frac{10\pi}{(4 + \pi)} \times \frac{4}{\pi}$$

$$2r = 5.601$$

$$r = 2.80 \text{ (3 sig fig)}$$

(d)

$$A = \frac{\pi}{4}(4 + \pi)(2.8)^2 - 10\pi(2.8) + 100$$

$$= 56$$

(e)

$$\frac{dA}{dr} = \frac{\pi}{4}(4 + \pi)(2r) - 10\pi$$

$$\frac{d^2A}{dr^2} = \frac{\pi}{4}(4 + \pi)(2)$$

Since $\frac{d^2A}{dr^2} > 0$, A is minimum.

(a)

$$\begin{aligned}
 \text{LHS:} \quad & \operatorname{cosec} 2\theta + \cot 2\theta \\
 &= \frac{1}{\sin 2\theta} + \frac{\cos 2\theta}{\sin 2\theta} \\
 &= \frac{1 + \cos 2\theta}{\sin 2\theta} \\
 &= \frac{2\cos^2 \theta}{2\sin \theta \cos \theta} \\
 &= \cot \theta
 \end{aligned}$$

(b)

$$\operatorname{cosec} 2\theta + \cot 2\theta + \frac{1}{\sqrt{3}} = 0$$

$$\operatorname{cosec} 2\theta + \cot 2\theta = -\frac{1}{\sqrt{3}}$$

$$\cot \theta = -\frac{1}{\sqrt{3}}$$

$$\tan \theta = -\sqrt{3}$$

$$\text{Basic angle} = \frac{\pi}{3}$$

$$\theta = \frac{2\pi}{3}, -\frac{\pi}{3}$$

(c)

$$f'(x) = \sin 2x + \cos 3x$$

$$f(x) = \frac{-\cos 2x}{2} + \frac{\sin 3x}{3} + c$$

$$f\left(\frac{\pi}{6}\right) = \frac{-\cos \frac{\pi}{3}}{2} + \frac{\sin \frac{\pi}{2}}{3} + c = 0$$

$$-\frac{1}{4} + \frac{1}{3} + c = 0$$

$$c = -\frac{1}{12}$$

$$f(x) = \frac{-\cos 2x}{2} + \frac{\sin 3x}{3} - \frac{1}{12}$$

$$f''(x) = 2\cos 2x - 3\sin 3x$$

$$f''(x) + 9f(x) = 2\cos 2x - 3\sin 3x + 9\left[\frac{-\cos 2x}{2} + \frac{\sin 3x}{3} - \frac{1}{12}\right]$$

$$= 2\cos 2x - 3\sin 3x - \frac{9}{2}\cos 2x + 3\sin 3x - \frac{9}{12}$$

$$= -\frac{3}{4} - \frac{5}{2}\cos 2x$$